SALESFORCE ADMIN COURSE • EPISODE 36

Field-Level Security & FLS Architecture

Protecting Sensitive Data at the Column Level: Read Access, Edit Access, FLS vs. Page Layouts, Dynamic Forms Limitations, Field Accessibility Diagnostics, and Practical Hands-on Testing.

- 🔒 Core Question: "Which fields can you see or edit?"
- 👁️ Read Access vs. 🖍️ Edit Access Mechanics
- ⚙️ FLS vs Page Layout: Security vs Presentation
- 🎛️ Dynamic Forms UI vs True Database Security
- 🔍 Field Accessibility Diagnostic Matrix
- 🧪 Hands-On Testing Lab with Employee__c Salary

Created by Sachin • Salesforce Certified Administrator Series • Duration: ~20–25 Mins

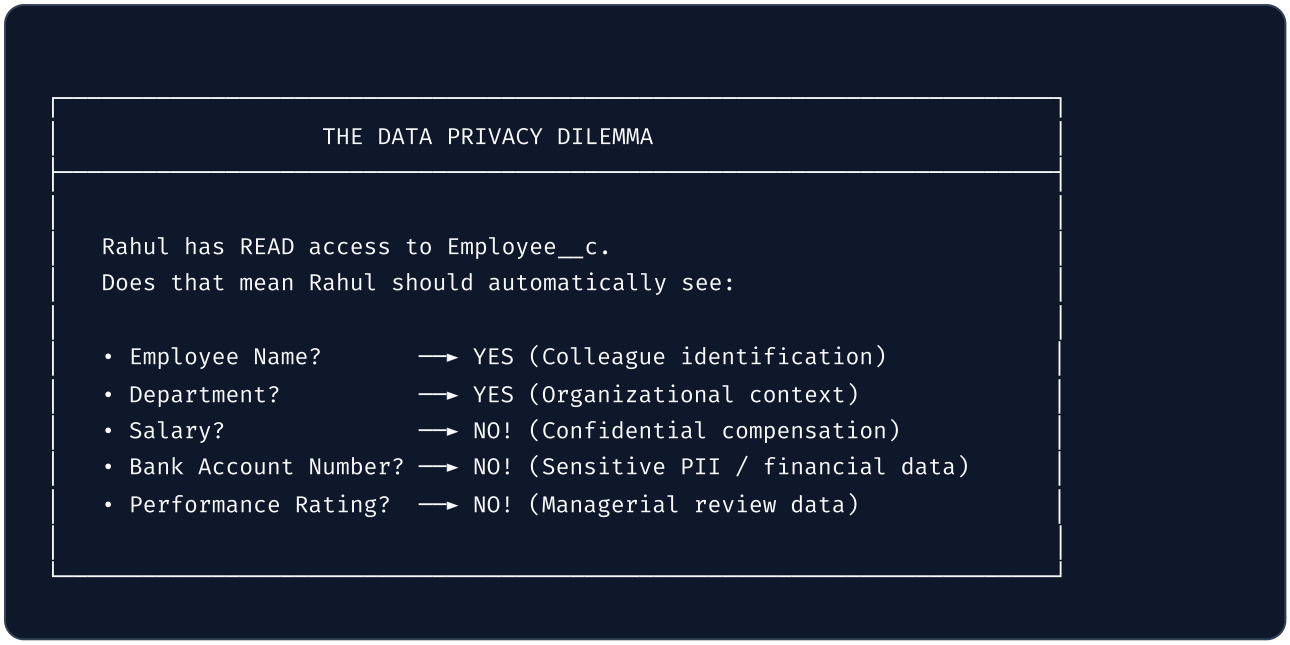
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1. Introduction: Zooming In from Object to Field

In the previous episode, we established **Object-Level Security** (CRUD). We answered: *"What can this user do with an entire object table?"*

However, granting a user access to an object introduces a major data privacy problem. Consider our custom object `Employee__c`:



Object-Level Security is a broad gatekeeper. It either lets a user into the room or keeps them out. It **cannot** selectively hide individual columns of data. This is precisely why Salesforce provides **Field-Level Security (FLS)**.

2. What is Field-Level Security (FLS)?

Field-Level Security (FLS) controls whether a user can see (read) or edit (write to) the value of a specific field on an object.

> *"Object-level security controls access to the entire record table; Field-level security controls access to individual fields (columns) within that record."*

When FLS hides a field from a user, Salesforce strips that field out completely across the **entire platform**:

- It is hidden from record detail and edit pages.
- It is hidden from List Views and Related Lists.
- It cannot be searched via Global Search or Lookup dialogues.
- It cannot be selected or viewed in Reports and Dashboards.
- It cannot be accessed or queried via the API, Apex (with security enforced), or Data Loader!

3. Read Access vs. Edit Access

Field-Level Security is configured using two fundamental permission checkboxes:

👁️ Read Access (Visible)

Controls whether the field is **visible** to the user. If unchecked, the user cannot see the field or its contents anywhere in Salesforce.

✏️ Edit Access (Read / Write)

Controls whether the user can **modify** the value of the field. If unchecked while Read is checked, the field becomes **Read-Only**.

The Absolute Dependency Rule of FLS:
You **CANNOT** grant Edit Access without granting Read Access! In the Salesforce setup interface, unchecking Read Access automatically unchecks and disables Edit Access.

READ ACCESS	EDIT ACCESS	USER EXPERIENCE	TYPICAL BUSINESS PERSONA
❌ Unchecked	❌ Unchecked	**Hidden Completely**	Standard Employees / Peers (cannot see Salary/Bank Details)
☑ Checked	❌ Unchecked	**Read-Only (Visible)**	Department Managers (can review Salary, but cannot change it)
☑ Checked	☑ Checked	**Read / Write (Editable)**	HR Administrators / Payroll Officers (full modification rights)
❌ Unchecked	☑ Checked	*Impossible State*	Salesforce UI prevents this combination automatically.

4. Where is FLS Configured?

Just like object permissions, field permissions follow our standard three-tier authorization hierarchy:

1. PROFILE FLS
Sets the mandatory baseline field visibility and read-only status for the user's role.

➕ combined with

2. PERMISSION SET FLS
Additively grants extra field visibility or editability without altering profiles.

📦 bundled into

3. PERMISSION SET GROUP
Bundles field permissions together across multiple permission sets by job function.

> **Additive Rule for FLS:**
> If a user's Profile has a field set to *Read-Only*, assigning a Permission Set with *Edit Access* checked makes the field **Editable** for that user!

5. Field Accessibility (The Diagnostic Matrix)

When users submit support tickets stating *"Why can't I edit this field?"* or *"Why is this field missing?"*, checking each profile one by one is slow.

Salesforce provides a dedicated administrator diagnostic tool called **Field Accessibility**:

```
Setup ➔ Security ➔ Field Accessibility (or Object Manager ➔ Object ➔ Fields ➔ Field Accessibility)
```

🔍 View by Fields

Select one specific field (e.g., `Salary__c`) to view a complete cross-matrix table showing its accessibility status across every Profile in your org.

👥 View by Profiles

Select a specific profile (e.g., `Sales User`) to view all fields on an object and see which ones are Editable, Read-Only, or Hidden.

> **Important Distiction:**
> Field Accessibility is a **diagnostic and inspection tool**. It helps administrators quickly analyze and modify field access across profiles and page layouts from one central grid—it is not an independent security layer.

6. FLS vs. Page Layout (The #1 Exam Trap)

This is one of the most critical concepts for any Salesforce Administrator. Confusing **Page Layout settings** with **Field-Level Security** leads to severe security vulnerabilities and failed certification questions!

CRITERIA	FIELD-LEVEL SECURITY (FLS)	PAGE LAYOUT SETTING
Primary Purpose	**True Database Security**	**User Interface & Presentation**
Where Configured?	Profiles & Permission Sets	Page Layout Editor (Object Manager)
Enforced in Reports?	YES — Always Hidden	NO — Visible in Reports!
Enforced in Search & Lookups?	YES — Fully Hidden	NO — Still Searchable!
Enforced in API & Data Loader?	YES — Blocked from API	NO — Read/Write via API!
Can it override the other?	**YES!** FLS is the ultimate authority.	**NO!** Cannot expose fields hidden by FLS.

The Golden Security Rule:
If you remove a field from a Page Layout, users **CAN STILL ACCESS IT** through Reports, List Views, Global Search, and the API / Data Loader! **Never rely on Page Layouts to protect sensitive data—ALWAYS use Field-Level Security!**

7. FLS vs. Dynamic Forms

With the introduction of Lightning **Dynamic Forms**, admins can conditionally show or hide fields on Lightning record pages based on filter criteria (e.g., *"Show 'Rejection Reason' only when Status = Closed Lost"*).

Field-Level Security (FLS)

Answers: **"Is this user authorized to see or edit this data?"**
Enforced globally at the data layer across UI, APIs, and reporting.

Dynamic Forms

Answers: **"Should this field appear on this specific screen right now?"**
Enforced only on the Lightning Page user experience.

Dynamic Forms Caution:
Just like Page Layouts, Dynamic Forms component visibility filters only adjust the page interface. A field hidden by Dynamic Forms criteria can still be queried via reports and APIs if FLS permits it.

8. Hands-On Practical Demo: Securing Salary__c

Follow these practical steps to implement and verify Field-Level Security on our custom `Employee__c` object:

STEP 01 — **Verify Fields on Custom Object Employee__c**

```
Setup ➔ Object Manager ➔ Employee ➔ Fields & Relationships
```

Confirm the presence of our fields: `Employee Name`, `Department__c`, `Salary__c` (Currency), `Bank_Account__c`, and `Performance_Rating__c`.

STEP 02 — **Configure Employee Persona (Rahul) ➔ Salary Hidden**

```
Setup ➔ Profiles ➔ [Sales User / Standard Employee Profile] ➔ Object Settings ➔ Employees
```

Scroll to **Field Permissions** and locate **Salary**:

- Read Access: Unchecked ❌
- Edit Access: Unchecked ❌

Click **Save**. The Salary field is now completely invisible to standard employees.

STEP 03 — **Configure Manager Persona (Amit / Priya) ➔ Salary Read-Only**

```
Setup ➔ Profiles (or Perm Set) ➔ [Manager Profile / Set] ➔ Object Settings ➔ Employees
```

Under **Field Permissions** for **Salary**:

- Read Access: Checked ✅
- Edit Access: Unchecked ❌

Click **Save**. Managers can review employee compensation for reviews, but cannot modify salary values.

STEP 04 **Configure HR / Payroll Persona ➔ Salary Read + Edit**

```
Setup ➔ Profiles (or Perm Set) ➔ [HR Profile / Set] ➔ Object Settings ➔ Employees
```

Under **Field Permissions** for **Salary**:

- Read Access: Checked ✅
- Edit Access: Checked ✅

Click **Save**. HR personnel have full read and write authority over Salary.

STEP 05 **Test All Three User Perspectives in the UI**

```
Setup ➔ Users ➔ Log in as each user ➔ Open the SAME Employee Record
```

Open identical record `EMP-001 (John Doe)` across each session:

- **Rahul (Employee):** Looks at the record. The **Salary field does not exist** on the page layout, list views, or reports!
- **Priya (Manager):** Opens the record. The **Salary field is visible ($85,000)**, but shows a lock icon 🔒 with no inline edit pencil.
- **HR Admin:** Opens the record. The **Salary field is visible ($85,000)**, and clicking the pencil icon allows modifying the value.

9. Multi-User Verification Matrix

USER PERSONA	OBJECT ACCESS	SALARY FLS	RECORD PAGE UI	REPORT VISIBILITY
Employee (Rahul)	Read + Create	Hidden ❌	Field is completely invisible	Cannot see column
Manager (Priya)	Read + Edit	Read-Only 👁️	Field visible; lock icon 🔒	Visible in reports
HR Officer (Admin / HR)	Full CRUD	Read + Edit ✏️	Field visible; editable ✏️	Visible in reports

10. Key Takeaways & Architectural Summary

1. FLS Operates at Column Level

While Object Security opens or closes the table, Field-Level Security decides which specific columns a user can see (Read) or edit (Write).

2. Read is Mandatory for Edit

You cannot grant Edit Access without Read Access. Hiding a field (Read = ❌) strips visibility across all UI, search, and APIs.

3. FLS Trumps Page Layouts

Page Layouts manage screen layout and UI preferences. If FLS restricts a field, putting it on a layout will NOT bypass security.

4. Never Secure via UI Tools

Neither Page Layouts nor Dynamic Forms are security tools. Sensitive data must always be protected using Field-Level Security.

11. Next Episode Preview

📺 Coming Up in Episode 37: Record-Level Security & Organization-Wide Defaults (OWD)

We have mastered **Object-Level Security** (the building door) and **Field-Level Security** (the data columns). Now we enter the deepest and most powerful layer of Salesforce security: **Record-Level Security**! In Episode 37, we answer: *"Both Rahul and Amit have access to the Employee object and can see all standard fields, but WHICH specific individual Employee records is Rahul allowed to see?"* We'll master **Organization-Wide Defaults (OWD)**, Public Read/Write, Public Read Only, and Private models!